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Basics of Machine Learning

- 3.1 Learning from examples
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Human Learning is the process of gaining information through observations.

In our day to day life, we perform several activities. To do the activity in a proper way, we must have proper information related to that activity. Also, we always try to improve our activity by gaining more information or by previous experience related to that one. So, with more learning, we can make our activity more efficient.

• Types of Human Learning :

1. Learning under expert guidance
2. Learning guided by knowledge gained from expert
3. Learning by self

1. Learning under expert guidance

A kid shows her head when we asked, "Where is your head?" Because, this thing she learns from her parents.

Guided learning is the process of gaining information from a person who has sufficient knowledge due to the past experience.

2. Learning guided by knowledge gained from experts

A kid can group together only those toys which belong to the group of "cars". Here, the kid has the only knowledge regarding which toy is a car. No one taught her to make a group of cars.

It is some past knowledge shared on some different way which is used as a learning to make decisions.

3. Learning by self :

The kids learn to walk through obstacles is the appropriate example for learning by self. Using past experience, observations; one can learn by self.

It is direct learning.

• Machine Learning :

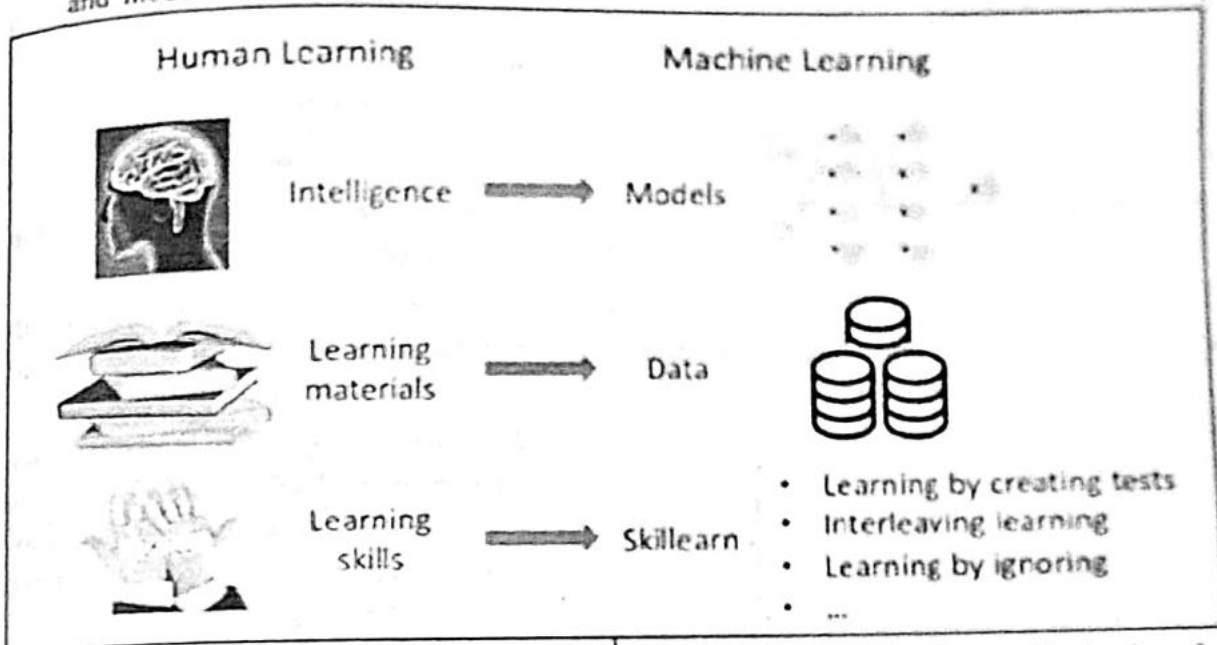
Machine Learning (ML) is basically that field of computer science with the help of which computer systems can provide sense to data in much the same way as human beings do.

Machine learning is a subfield of Artificial Intelligence (AI) which extracts patterns

out of raw data by using an appropriate algorithm.

Its goal is to enable computers to learn on their own from experience.

A machine's learning algorithm enables it to identify patterns in observed data, build models that explain the world, and predict things without having explicit pre-programmed rules and models.



- Machine Learning (ML) is a field of AI consisting of learning algorithms that improve their performance (P) at executing some task (T) over time with experience (E).

Definition:

As per Tom M. Mitchell, Machine Learning is defined as, "A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P, if its performance at tasks in T, as measured by P, improves with experience E."

For example, Robot Driving in a maze.

- Class of task: To reach at the end point
- Performance Measure: Time taken to reach at the end point

– Source of experience: Navigating the maze from start to end by the robot

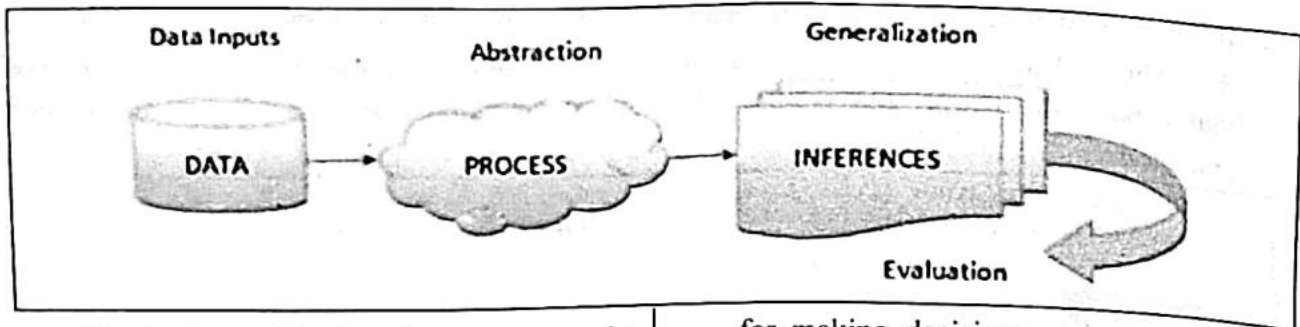
Remember that,

Task (T): It is a real world problem. Classification, clustering, regression are the examples of ML based task.

Experience (E): It is the knowledge gained from data points provided to the algorithm or model. The experience gained by our ML model or algorithm will be used to solve the task T.

Performance (P): The measure which tells whether ML algorithm is performing as per expectation or not is its performance. An ML algorithm is supposed to perform task and gain experience with the defined time period.

How do machines learn?



The basic machine learning process can be divided in four parts:

1. **Data Inputs** : Past data is used as a base for future prediction.

The vast pool of knowledge is available from the input data. The machine can use the past data for decision-making process.

2. **Abstraction** : The input data is represented in a broader way using appropriate algorithms.

Abstraction helps to derive a model which is a summarized knowledge representation of the raw data. The model may be a mathematical equation, rule-based computational blocks, logical combinations of same observations or well-defined data structures.

The decision related to choice of a model is based on the type of a problem, nature of input data, domain of the problem, and many more.

In this part, we get the finite set of data which is known as **training data set**.

Once model is chosen then the next step is to fit the model based on the input received.

3. **Generalization** : The abstracted representation is generalized to create a framework

for making decisions.

Generalization is the most critical part in ML process.

In this part, the abstracted knowledge will be tuned up to a form which can be used to take decisions.

We want to apply the model to take decision on a set of unknown data which is called **test data set**. For decision-making, approximate/heuristic approach has to be taken.

3. **Evaluation** : Evaluation is the process of giving feedback to the user to measure the utility of the learned knowledge. This feedback is then utilized to effect improvements in the whole learning process.

3.1 Learning from Examples

Learning from examples is a core concept in machine learning, particularly within the realm of supervised learning. It involves training algorithms to make predictions or decisions based on input data paired with known outcomes. Here's a detailed explanation of the process:

- | | |
|-------------------|-------------------|
| 1. Definition | 2. Key Components |
| 3. Generalization | 4. Evaluation |

1. Definition

Learning from examples refers to the ability of a machine learning model to learn patterns and relationships in data by using labelled training examples. Each example consists of features (input variables) and corresponding labels (output values or classes).

2. Key Components

- **Training Data** : A collection of examples used to train the model. Each example typically includes a set of features and a label. For instance, in a dataset of emails, features could include the presence of certain words, while labels might indicate whether an email is "spam" or "not spam."
- **Model** : The mathematical representation that learns from the training data. Common models include decision trees, neural networks, and support vector machines.
- **Training Process** : The model adjusts its parameters based on the training data to minimize errors in its predictions. This is often done using algorithms like gradient descent, which iteratively refine the model to improve accuracy.

3. Generalization

A crucial goal of learning from examples is generalization—ensuring that the model performs well on unseen data, not just the training set. Overfitting occurs when a model learns the training data too well, capturing noise rather than the underlying pattern, which can lead to poor performance on new examples.

4. Evaluation

After training, the model is evaluated using a separate test dataset, which was not part of the training process. This evaluation helps assess the model's ability to generalize. Common evaluation metrics include:

- **Accuracy** : The proportion of correctly predicted instances.
- **Precision** : The ratio of true positive predictions to the total predicted positives.
- **Recall** : The ratio of true positive predictions to the total actual positives.

Applications :

Learning from examples is widely applicable in various fields :

- **Image Recognition** : Classifying images based on labelled datasets (e.g., identifying cats vs. dogs).
- **Natural Language Processing** : Tasks such as sentiment analysis, where models learn to classify text based on labelled sentiments (positive, negative).
- **Medical Diagnosis** : Using patient data to predict diseases based on historical examples.

Learning from examples is a powerful approach that enables machines to improve their performance through experience. By leveraging historical data, models can learn to make informed predictions, adapting to new information and scenarios. This capability underlies many modern applications of artificial intelligence and data science.

- **Well-posed learning problem** :

A (machine learning) problem is well-

posed if a solution to it exists, if that solution is unique, and if that solution depends on the data / experience but it is not sensitive to (reasonably small) changes in the data / experience.

To have a well-defined learning problem, three features needs to be identified:

1. The class of tasks (T)
2. The measure of performance to be improved (P)
3. The source of experience (E)

Example, Face Recognition Problem

- Class of Task: predicting different types of faces
- Performance Measure: able to predict maximum types of faces
- Source of Experience: training machine with maximum amount of datasets of different face images

To solve well-posed problem using Machine Learning, a simple framework this involves three questions to answer:

1. What is the problem? (Formalism the problem based on Task, Experience and Performance)
2. Why does the problem need to be solved? (Motivation and benefits to solve the problem)
3. How to solve the problem ? (Creating the model/algorithm using knowledge)

• TYPES OF DATA:

Data (Attributes) are broadly classified into Qualitative Data and Quantitative Data.

- A) Qualitative Data
B) Quantitative Data

A) Qualitative Data :

The data collected on grounds of categorical variables are **qualitative data**. It is also known as categorical data.

Qualitative data are more descriptive and conceptual in nature.

It measures the data on basis of the type of data, collection, or category.

Qualitative data talks about the experience or quality and explains the questions like 'why' and 'how'.

For example, if we consider the performance of employee in terms of Good, Average or Poor then it is called qualitative data.

The qualitative data are subdivided into two parts: Nominal Data and Ordinal Data.

Nominal data is the data which has no numeric value, but a named value.

Example: Nationality: Indian, American, British, etc.

On nominal data, mathematical functions like addition, multiplication or statistical functions like mean, median cannot be applied. Only basic count should be applied. So we can get mode of the nominal data.

Ordinal data is the data which has named value which can be naturally ordered in increasing or decreasing.

Example : Students Satisfaction: Very Happy, Happy, Unhappy

On ordinal data, the ordered data is available so we can perform mode, median and quartiles operations. But, we cannot perform operations like mean.

B) Quantitative Data :

The data collected on the grounds of the numerical variables are **quantitative data**. It is also known as numeric data.

Quantitative data are more objective and conclusive in nature.

It measures the values and is expressed in numbers. The data collection is based on "how much" is the quantity.

Quantitative data talks about the quantity and explains the questions like 'how much', 'how many'.

For example, if we consider marks of Mid Semester Exam like '30', '25', '12' then

it is called quantitative data.

The quantitative data are subdivided into two parts: Interval Data and Ratio Data.

Interval Data is numeric data for which exact difference between values is known. Such data do not have something called a 'true zero' value.

Example : Celsius Temperature, Date, Time

Ratio Data is numeric data for which exact value can be measured.

Such data have the 'true zero' value, too.

Example: Height, Weight, Age, Marks

3.2 Forms of Learning -Supervised learning, Unsupervised learning, Reinforcement learning

Supervised Learning

- > Labeled data
- > Direct Feedback
- > Predict outcome/Future

Unsupervised Learning

- > No Labels
- > No Feedback
- > Fine hidden Structure in data

Reinforcement Learning

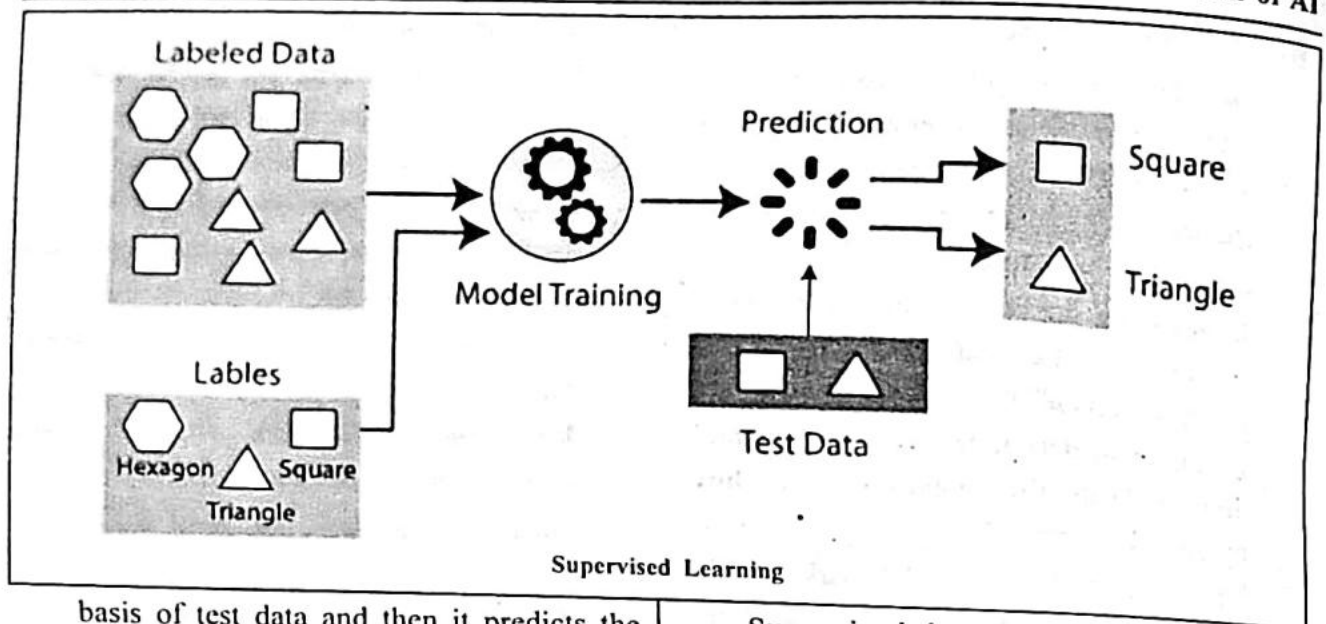
- > Decision process
- > Reward system
- > Learn series of actions

3.2.1 Supervised Learning :

Supervised learning is the types of machine learning in which machines are trained using well "labelled" training data, and on basis of that data, machines predict the output.

The labelled data means some input data is already tagged with the correct output.

In supervised learning, models are trained using labelled dataset, where the model learns about each type of data. Once the training process is completed, the model is tested on the



basis of test data and then it predicts the output.

Supervised learning is a process of providing input data as well as correct output data to the machine learning model. The aim of a supervised learning algorithm is to find a mapping function to map the input variable(x) with the output variable(y).

In the real-world, supervised learning can be used for Risk Assessment, image classification, Fraud Detection, spam filtering, etc.

Regression and classification are two main areas of Supervised Learning.

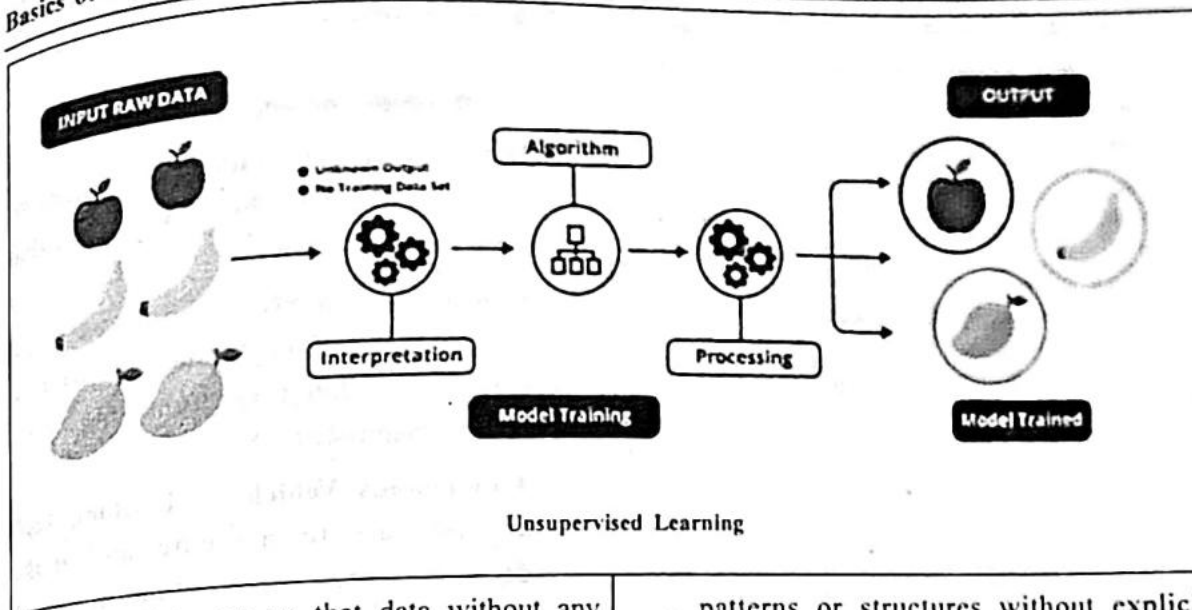
Regression algorithms are used if there is a relationship between the input variable and the output variable. It is used for the prediction of continuous variables, such as Weather forecasting, Market Trends, etc. Classification algorithms are used when the output variable is categorical, which means there are two classes such as Yes-No, Male-Female, True-false, etc.

Supervised learning involves training models on labelled datasets, where each input is associated with a correct output. Applications include:

1. Image Classification: Identifying objects within images (e.g., classifying images of cats vs. dogs).
2. Spam Detection: Filtering email messages as "spam" or "not spam" based on labelled examples.
3. Sentiment Analysis: Classifying text (e.g., reviews or social media posts) as positive, negative, or neutral.
4. Medical Diagnosis: Predicting diseases based on patient data and historical cases (e.g., detecting tumours in medical images).
5. Credit Scoring: Assessing the likelihood of a borrower defaulting on a loan using historical financial data.

3.2.2 Unsupervised Learning

Unsupervised learning is a type of machine learning in which models are trained using unlabelled dataset and are



allowed to act on that data without any supervision.

The main aim of the unsupervised learning algorithm is to group or categories the unsorted dataset according to the similarities, patterns, and differences.

In the real world, unsupervised learning can be used for pattern recognition, inventory management, data exploration, recommender system, etc.

Unsupervised learning is classified into two categories of algorithms:

- **Clustering:** A clustering problem is where you want to discover the inherent groupings in the data, such as grouping customers by purchasing behaviour.
- **Association:** An association rule learning problem is where you want to discover rules that describe large portions of your data, such as people that buy X also tend to buy Y.

Unsupervised learning deals with unlabelled data, where the model identifies

patterns or structures without explicit guidance.

• **Applications include :**

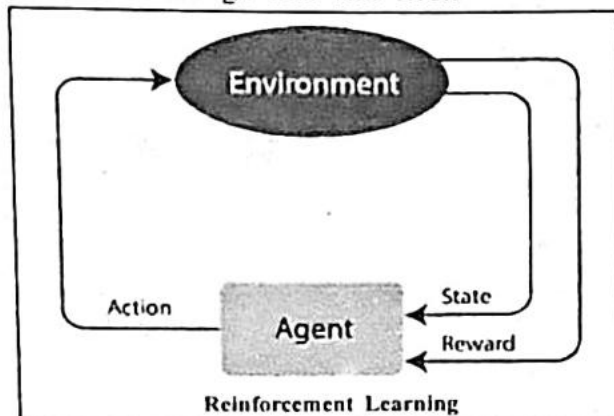
1. **Clustering :** Grouping similar data points, such as customer segmentation in marketing to identify distinct customer profiles.
2. **Anomaly Detection :** Identifying unusual patterns in data, which can be useful in fraud detection or network security.
3. **Dimensionality Reduction :** Simplifying data while preserving essential features (e.g., using PCA for visualization or improving model performance).
4. **Recommendation Systems :** Suggesting products or content based on user behavior and preferences (e.g., Netflix or Amazon recommendations).
5. **Market Basket Analysis :** Discovering associations between products purchased together to optimize sales strategies.

3.2.3 Reinforcement Learning:

Reinforcement learning is a type of machine learning method where an

intelligent agent (computer program) interacts with the environment and learns to act within that.

In general, a reinforcement learning agent is able to perceive and interpret its environment, take actions and learn through trial and error.



The Reinforcement Learning problem involves an agent exploring an unknown environment to achieve a goal. It is based on the hypothesis that all goals can be described by the maximization of expected cumulative reward. The agent must learn to sense and perturb the state of the environment using its actions to derive maximal reward.

There are four elements of reinforcement learning: policy, reward function, value function and model of the environment.

A reward function is a function that provides a numerical score based on the state of the environment. If it makes the right decision, it gets a reward. If it makes mistakes, it receives a penalty.

In the real world, reinforcement learning can be used for self-driving cars, gaming, robotics, etc.

Reinforcement learning involves training agents to make decisions by taking actions

in an environment to maximize cumulative rewards.

• **Applications include :**

1. **Game Playing :** Training AI agents to play games like Chess, Go, or video games (e.g., AlphaGo).
2. **Robotics :** Enabling robots to learn complex tasks through trial and error in real or simulated environments (e.g., robotic manipulation).
3. **Autonomous Vehicles :** Teaching self-driving cars to navigate and make decisions based on real-time feedback from their environment.
4. **Dynamic Pricing :** Adjusting prices in real-time based on market demand and competition, maximizing profit through learned strategies.
5. **Personalized Learning :** Developing educational platforms that adapt to individual student needs based on their interactions and progress.

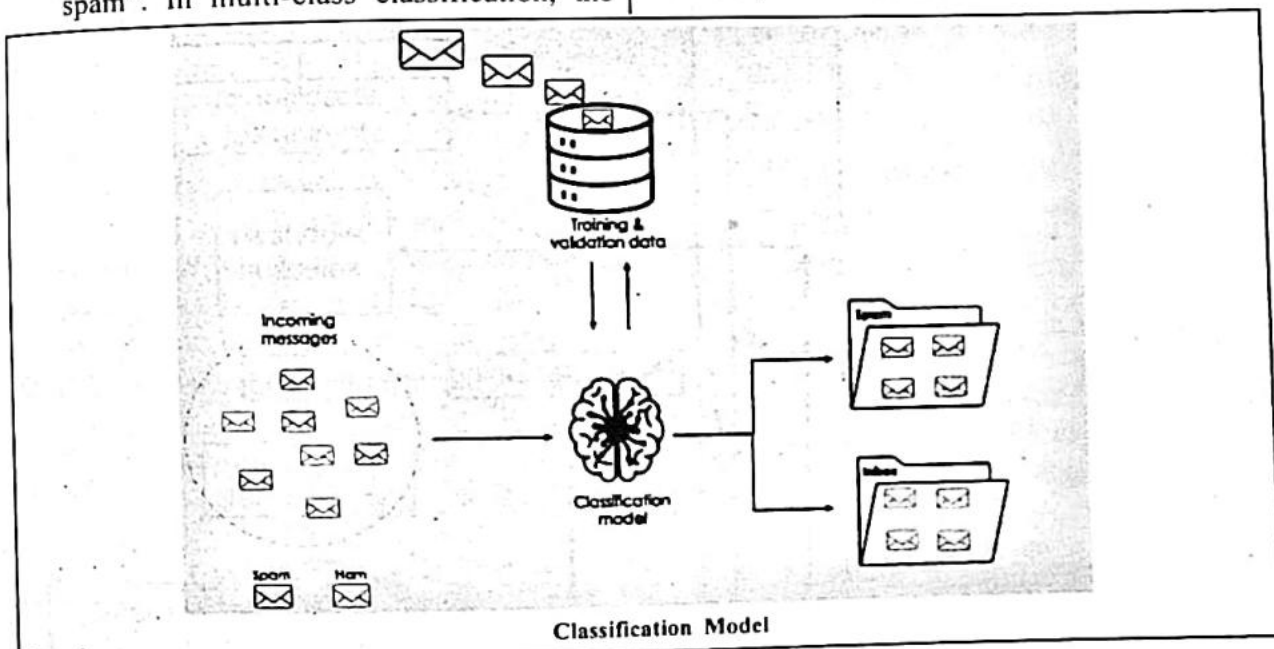
3.2.4 Difference between Supervised Learning, Unsupervised Learning and Reinforcement Learning

Classification is a type of supervised learning where the goal is to predict the categorical output variable for a given set of input features. In classification, the model learns to distinguish between different classes based on labelled data. Classification is a type of supervised learning where the goal is to predict the categorical output variable for a given set of input features. In classification, the model learns to distinguish between

Criteria	Supervised ML	Unsupervised ML	Reinforcement ML
Definition	Machine Learns by using labelled data	Machine is trained using unlabelled data without any guidance.	Agent interacts with the environment by performing action. Learns by errors and rewards.
Type of data	Labelled data	Unlabelled data	No – predefined data
Type of problems	Regression and classification	Association and Clustering	Reward and error based.
Supervision	External supervision	No supervision	No supervision
Algorithms	Linear Regression, Logistic Regression, Naive Bayes, Decision trees	K – Means clustering, KNN (K-nearest neighbours), Principle Component Analysis, Neural Networks	Monte Carlo, Q-Learning, SARSA
Aim	Calculate outcomes	Discover underlying patterns	Learn a series of action
Approach	Maps labelled inputs to the known outputs	Understands patterns & discover the output	Follow the trial and error method
Application	Risk Evaluation, Forecast Sales	Recommendation System, Anomaly Detection	Self-Driving Cars, Gaming, Healthcare

different classes based on labeled data. For example, in a binary classification problem, the goal is to predict one of two possible classes, such as “spam” or “not spam”. In multi-class classification, the

goal is to predict one of several possible classes, such as identifying different types of animals from an image. To build a classification model, the first step is to collect labeled data. This



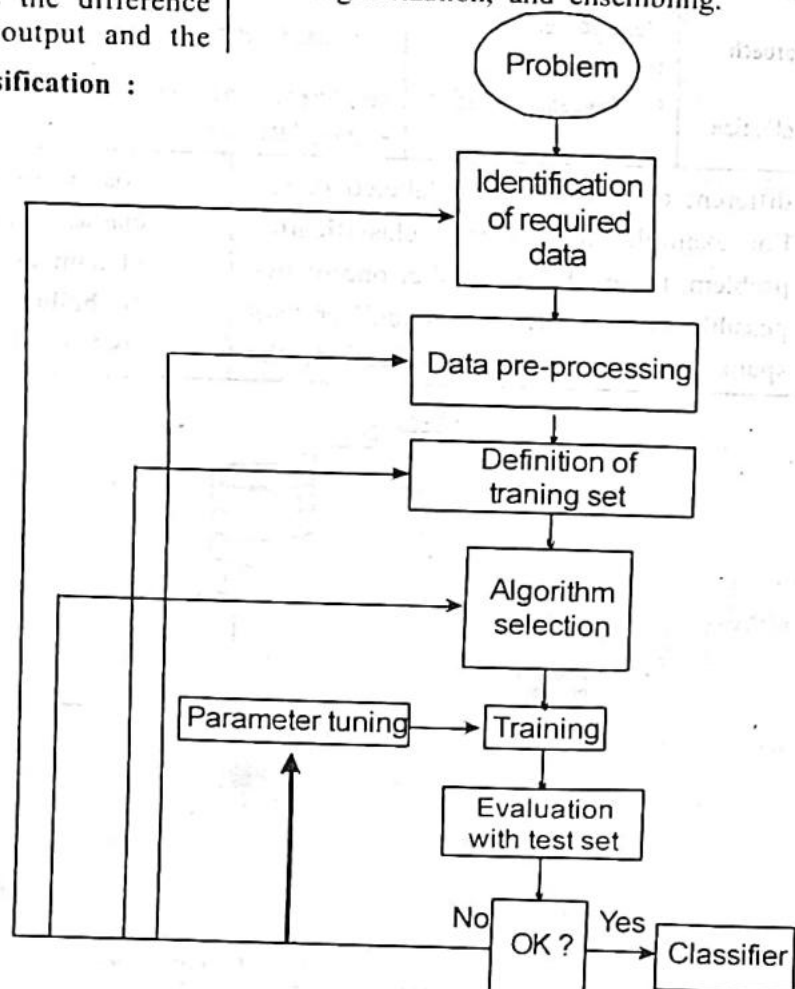
involves creating a dataset where each data point includes a set of input features and a corresponding output label. The input features can be anything that might be relevant to the classification task, such as pixel values for image data or demographic data for predicting customer behavior.

Once the dataset is prepared, the next step is to choose an appropriate classification algorithm, such as logistic regression, decision trees, or neural networks. The algorithm is then trained on the labeled dataset to learn the relationship between the input features and the output labels. During training, the algorithm adjusts its parameters to minimize the difference between the predicted output and the

actual output in the labeled dataset. Once the model is trained, it can be used to predict the output label for new input data. In summary, classification is a type of supervised learning where a target feature, which is of categorical type, is predicted for test data on the basis of the information imparted by the training data. The target categorical feature is known as class.

Evaluation of a classification model involves measuring its accuracy on a held-out validation dataset or through cross-validation techniques. The accuracy of the model can be improved by using techniques such as feature engineering, regularization, and ensembling.

- **Learning Steps of Classification :**



Step 1: Problem Identification: Identifying the problem is the first step in the supervised learning model. The problem must be well-posed problem.

Step 2: Identification of Required Data: Collecting and preparing a labeled dataset that includes input features and corresponding output labels.

Step 3: Data Preprocessing: Cleaning and preparing dataset, which may involve removing missing data, handling outliers and transferring the data into a suitable format.

Step 4: Definition of Training Data Set: Before starting the analysis, the user should decide type of data set is to be used as a training set. The training set needs to be actively representative of the real-world use of the given problem.

Step 5: Algorithm Selection: Choosing an appropriate classification algorithm based on the nature of the problem and the size and complexity of the dataset.

Step 6: Training: Using the labeled dataset to train the selected algorithm by adjusting its parameters to minimize the difference between the predicted output and the actual output. Also, adjusting the hyperparameters of the algorithm to optimize the model's performance on the validation dataset should be done, if it needs..

Step 7: Evaluation with the test dataset: Training data is run on the algorithm, and its performance is measured here. If a desired output is not obtained, again training of parameters may be required.

Regression in supervised learning is a type of predictive modelling that involves

estimating a continuous numerical output value based on input features. The goal of regression is to find a mathematical relationship between the input variables (also called predictors or independent variables) and the output variable (also called the response or dependent variable). The output value in regression can be any numerical value, including integers, real numbers, or even negative values. Regression models can be used for both linear and nonlinear relationships between the input and output variables.

The main goal of regression is to create a function that can predict the output variable for new inputs with a certain degree of accuracy. In order to achieve this goal, regression algorithms use statistical techniques to find the optimal parameters of the model that minimize the difference between the predicted values and the actual values in the training dataset.

• Applications of Regression :

- Predicting sales or revenue based on marketing spend or other factors.
- Estimating the price of a house based on features such as location, size, and number of rooms.
- Forecasting demand for a product based on historical sales data.
- Predicting patient outcomes based on medical data.
- Analyzing trends in financial data such as stock prices or interest rates.

3.3 Simple Models –Linear regression, Logistic regression, Support Vector Machines (SVM)

3.3.1 Simple Linear Regression

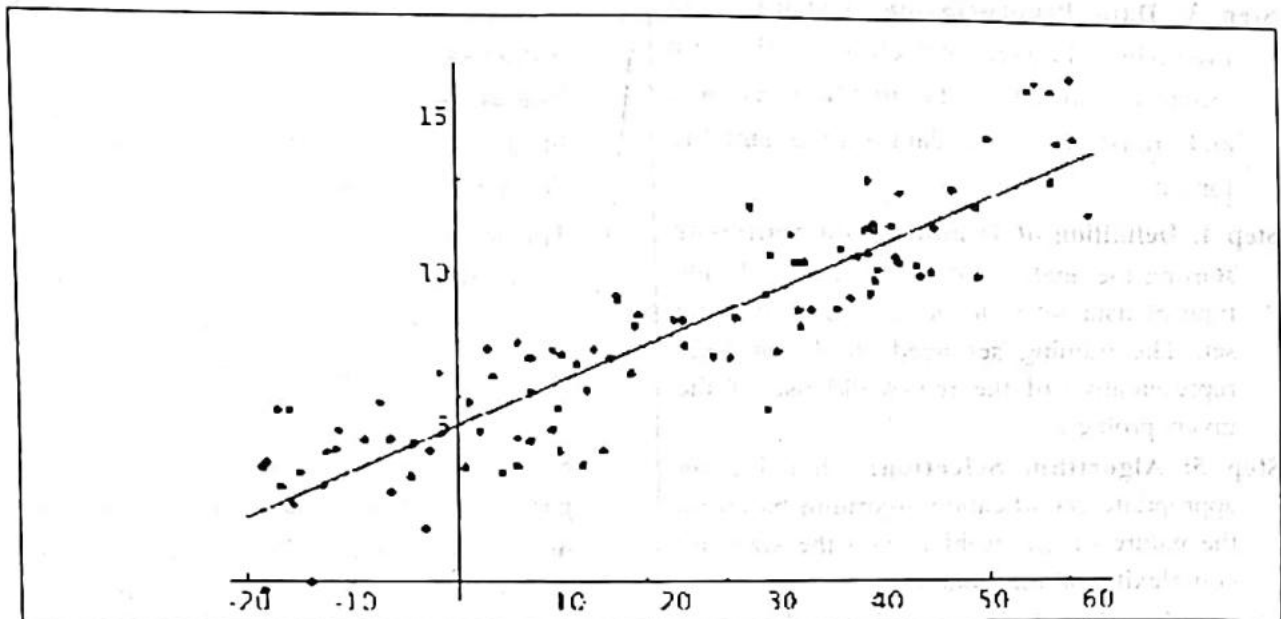
Linear regression is one of the most basic types of regression in machine learning. The linear regression model consists of a predictor variable and a dependent variable related linearly to each other. In case the

data involves more than one independent variable, then linear regression is called multiple linear regression models.

The below-given equation is used to denote the linear regression model:

$$y = mx + c + e$$

where m is the slope of the line, c is an intercept, and e represents the error in the model.



Simple linear regression is a type of regression analysis that is used to model the relationship between two variables, where one variable is considered the predictor or independent variable, and the other is the response or dependent variable. The goal of simple linear regression is to find the best linear relationship between the two variables.

Here is Suppose we have a dataset of 100 homes with two variables: square footage (independent variable) and sale price (dependent variable). We want to determine the relationship between these two variables, and specifically, how much

the sale price of a home increases for each additional square foot of living space.

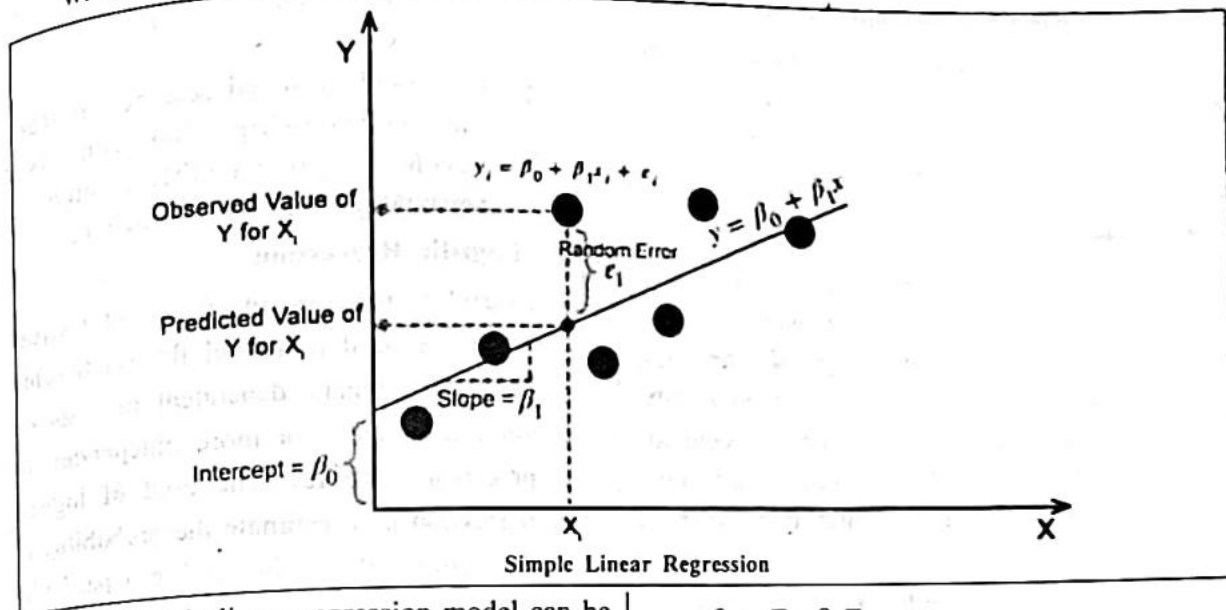
We can use simple linear regression to estimate the line of best fit for the data, which is a straight line that minimizes the sum of the squared differences between the predicted and actual values.

In simple linear regression, the model is expressed as:

$$Y = \beta_0 + \beta_1 X + \epsilon$$

where Y is the dependent variable, X is the independent variable, β_0 is the intercept, β_1 is the slope associated with the predictor variable X , and ϵ is the

random error term. The slope β_1 indicates the change in the dependent variable associated with a one-unit change in the independent variable X .



The simple linear regression model can be fitted to the data using various techniques such as ordinary least squares (OLS) or maximum likelihood estimation (MLE). The fitted model can be used to make predictions for new values of the predictor variable or to test hypotheses about the relationship between the variables.

The validity of the simple linear regression model depends on certain assumptions such as linearity, normality, homoscedasticity, and independence of the error term. Violations of these assumptions can lead to biased and inefficient estimates and incorrect inference.

To estimate the values of β_0 and β_1 , we can use the least squares method, which involves minimizing the sum of the squared errors between the predicted and actual values of the dependent variable. The equation for the least squares estimates of β_0 and β_1 are:

$$\beta_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$\beta_0 = \bar{y} - \beta_1 \bar{x}$$

Where x_i and y_i are the values of the independent and dependent variables for the i^{th} observation, $\bar{x}$ and $\bar{y}$ are the mean values of the independent and dependent variables, and the sums are taken over all observations in the dataset.

Once we have estimated the values of β_0 and β_1 , we can use the equation of the line to predict the sale price of a home based on its square footage.

For example, if the estimated equation of the line is:

$$y = 100000 + 100x$$

This means that for every additional square foot of living space, the sale price of the home is expected to increase by Rs. 100. So, a home with 1500 square feet of living space would be expected to sell for Rs. 250,000, $(100000 + 100 \cdot 1500)$.

• Pros :

- Simple linear regression is easy to implement and interpret.

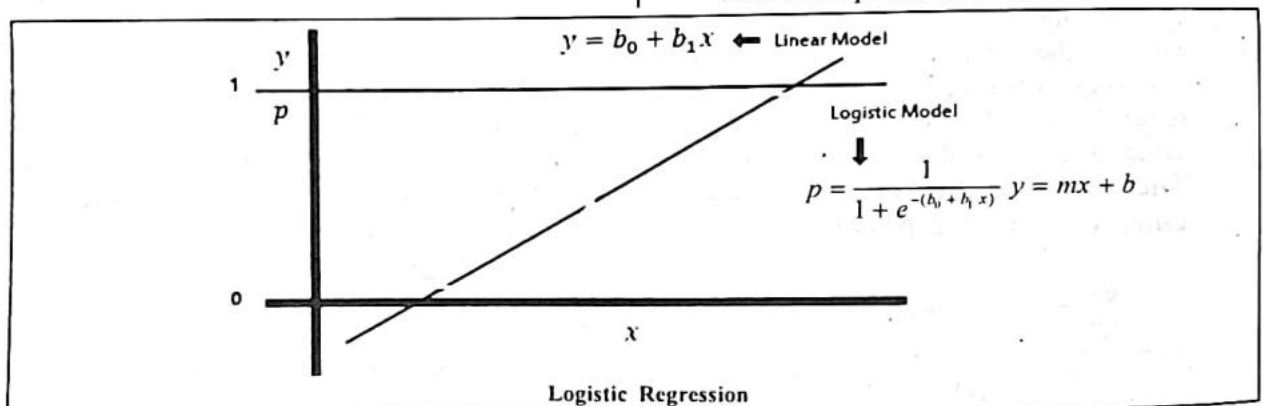
- It provides a simple way to quantify the strength and direction of the relationship between two variables.
- It can be used to make predictions and estimate the value of the response variable for a given value of the predictor variable.
- **Cons :**
 - Simple linear regression assumes that there is a linear relationship between the predictor and response variables, which may not be the case in reality.
 - It assumes that the errors or residuals are normally distributed and have constant variance, which may not hold in practice.
 - It can be sensitive to outliers, influential observations, and violations of the underlying assumptions.
- **Applications :**
 - Simple linear regression is commonly used in finance to model the relationship between two financial variables, such as stock prices and interest rates.
 - It is used in marketing to analyze the impact of advertising on sales or the relationship between price and demand for a product.

- It is used in engineering to model the relationship between two physical variables, such as temperature and pressure.
- It is used in social sciences to study the relationship between two psychological variables, such as personality and job satisfaction.

3.3.2 Logistic Regression

Logistic regression is a statistical technique used to model the relationship between a binary dependent or response variable and one or more independent or predictor variables. The goal of logistic regression is to estimate the probability of the dependent variable taking a particular value (e.g., 1 or 0) given the values of the predictor variables.

The logistic regression model assumes that the relationship between the predictor variables and the log-odds of the dependent variable is linear. The model can be fitted to the data using various techniques such as maximum likelihood estimation (MLE) or Bayesian methods. The fitted model can be used to make predictions for new values of the predictor variables or to test hypotheses about the relationship between the variables.



In above formula of the logistic regression the constant (b_0) moves the curve left and right and the slope (b_1) defines the steepness of the curve.

• **Types of Logistic Regression :**

1. Binary Logistic Regression
2. Multinomial Logistic Regression
3. Ordinal Logistic Regression

1. **Binary Logistic Regression :**

The categorical response has only two possible outcomes. Example : Spam or Not.

2. **Multinomial Logistic Regression :**

Three or more categories without ordering. Example: Predicting which food is preferred more (Veg, Non-Veg, Vegan).

3. **Ordinal Logistic Regression :**

Three or more categories with ordering. Example: Movie rating from 1 to 5

• **Pros of logistic regression:**

- Logistic regression is a powerful and widely used statistical technique for binary and ordinal classification problems, and can handle multiple predictor variables.
- The output of logistic regression is easy to interpret as it provides the predicted probabilities of the dependent variable taking a particular value.
- Logistic regression can handle both categorical and continuous predictor variables and can detect interactions between them.
- Logistic regression can be used to test hypotheses about the relationships between the variables and to estimate the impact of the predictor variables on the dependent variable.

• **Cons of logistic regression :**

- Logistic regression assumes that the relationship between the predictor variables and the dependent variable is linear, which may not hold in all cases.
- Logistic regression assumes that the observations are independent, which may not hold in cases where the data are clustered or correlated.
- Logistic regression can be sensitive to outliers and influential observations, which may affect the estimated coefficients and the goodness-of-fit of the model.
- Logistic regression may suffer from overfitting or underfitting if the model is too complex or too simple, respectively.

• **Applications of logistic regression :**

- Logistic regression is widely used in various fields such as healthcare, marketing, finance, and social sciences for binary and ordinal classification problems such as disease diagnosis, customer churn, credit scoring, and voter prediction.
- Logistic regression is used in biomedical research to model the probability of a disease occurring based on risk factors such as age, sex, and lifestyle factors.
- Logistic regression is used in social sciences to model the likelihood of a particular behaviour or attitude given demographic and psychographic characteristics of the individuals.
- Logistic regression is used in marketing to model the likelihood of a customer buying a product given their past behaviour and demographic characteristics.

Future	Simple Linear Regression	Logistic Regression
Purpose	Predicts a continuous dependent variable	Predicts a categorical dependent variable (binary)
Output	Continuous value (e.g., salary, temperature)	Probability of class membership (e.g., yes/no)
Relationship	Assumes a linear relationship	Assumes a linear relationship between predictors and log odds
Dependent Variable Type	Continuous	Categorical (typically binary)
Error Measurement	Mean Squared Error (MSE)	Accuracy, precision, recall, AUC-ROC
Assumptions	Linearity, homoscedasticity, normality of errors	Linear relationship in log odds, no multicollinearity
Use Cases	Predicting quantities (e.g., sales, height)	Classifying categories (e.g., spam detection)

Here's a comparison between simple linear regression and logistic regression:

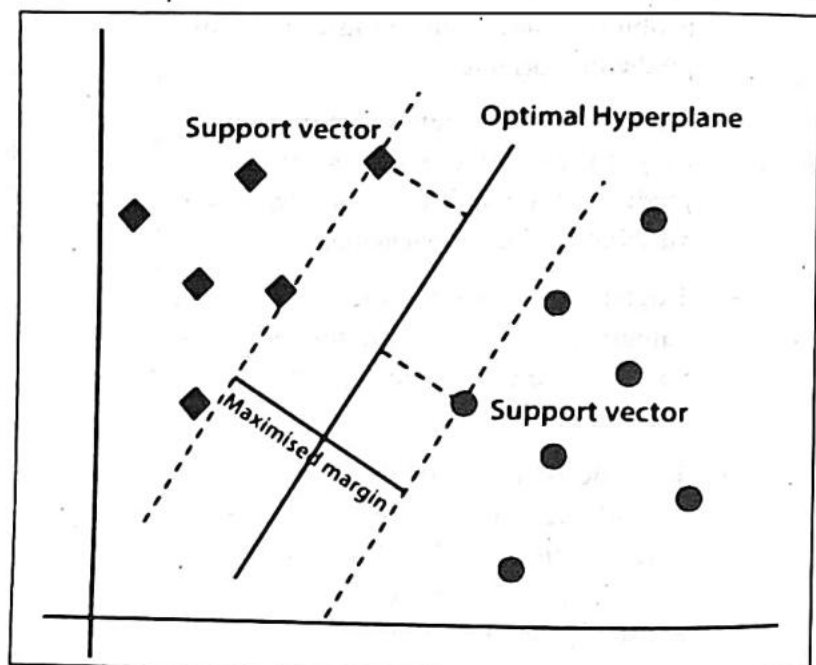
3.3.3 Support Vector Machines (SVM) :

Support Vector Machine (SVM) is a powerful classification algorithm used in supervised learning. The algorithm works by finding the optimal hyperplane that separates the different classes in the input data.

Support vectors are the data points (representing classes), the critical component in a data set, which are near the identified set of lines (**hyperplane**). In the case of a binary classification problem, SVM aims to find the hyperplane that maximizes the margin between the two classes. The **margin** is defined as the distance between the hyperplane and the closest data points from each class. SVM finds the optimal hyperplane by solving a convex optimization problem

that maximizes the margin while minimizing the classification error.

If the input data is not linearly separable, SVM can still be used by applying a kernel function that maps the data into a higher-dimensional feature space, where a linear hyperplane can be used to separate the classes. Some popular kernel functions used in SVM include linear, polynomial,



radial basis function (RBF), and sigmoid.

• SVM has several advantages, including :

- Effective for high-dimensional data.
- Can handle non-linearly separable data by applying kernel functions.
- Works well with small to medium-sized datasets.
- Robust to outliers.

However, SVM also has some limitations, including :

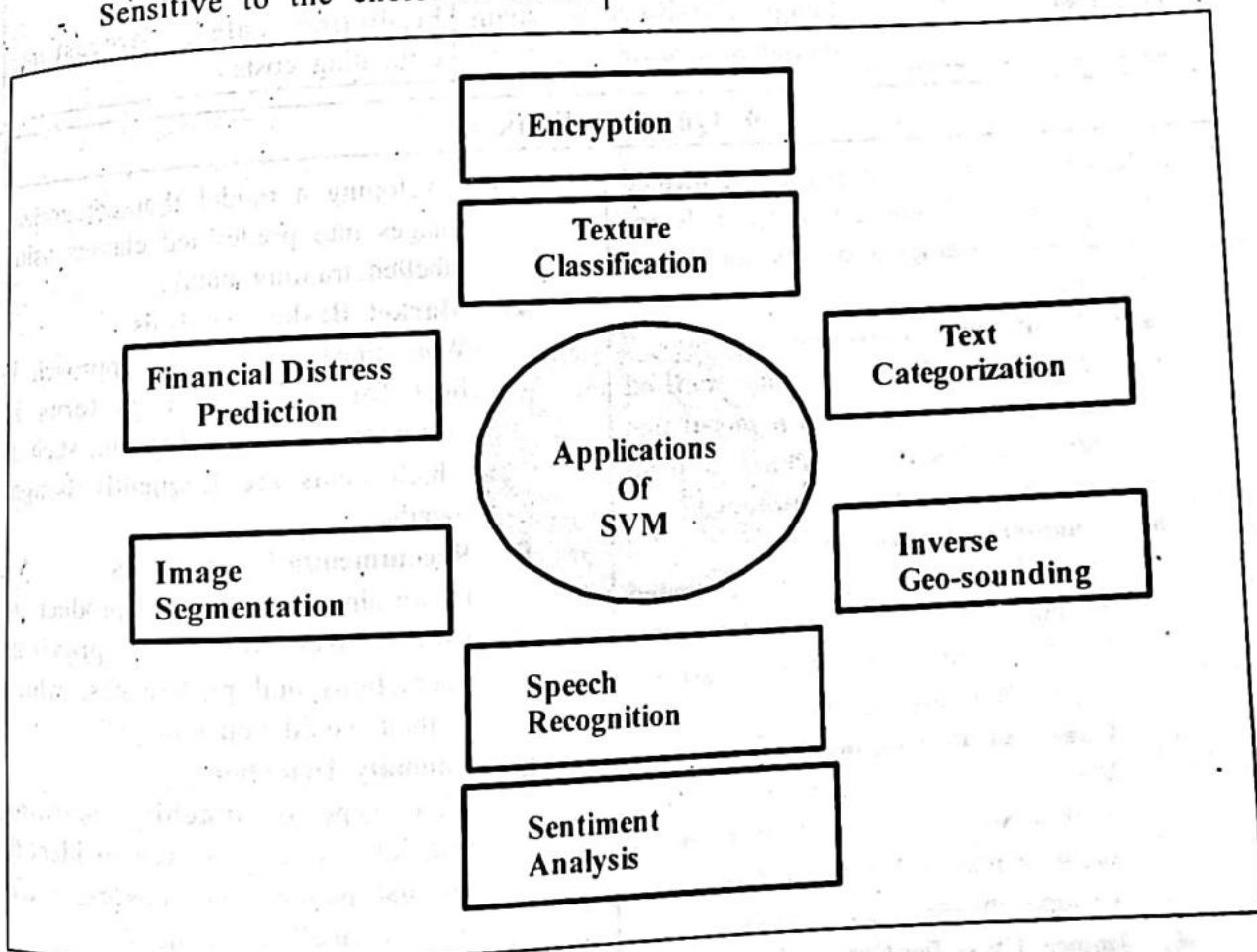
- SVM is applicable only for binary classification.
- Computationally expensive for large datasets.
- Sensitive to the choice of kernel

function and parameters.

- Difficult to interpret the resulting model and extract insights from it.
- Overfitting may cause.

• Applications of SVM include :

- Image classification and object recognition.
- Text classification and sentiment analysis.
- Bioinformatics and gene expression analysis.
- Financial forecasting and risk management.
- Anomaly detection in network intrusion detection systems.



• **Difference between Classification and Regression:**

Here's a comparison between classification and regression:

Feature	Classification	Regression
Purpose	Assigns input data to discrete categories	Predicts a continuous output value
Output	Categorical labels (e.g., spam or not spam)	Continuous values (e.g., price, temperature)
Examples of Algorithms	Decision Trees, Random Forest, SVM, Logistic Regression	Linear Regression, Polynomial Regression, SVR
Performance Metrics	Accuracy, Precision, Recall, F1 Score, AUC-ROC	Mean Squared Error (MSE), R-squared, MAE
Type of Target Variable	Categorical (nominal or ordinal)	Continuous
Data Distribution	Focuses on probability distributions for classes	Assumes a linear or non-linear relationship
Use Cases	Image classification, spam detection, medical diagnosis	Predicting sales, forecasting, estimating costs

❖ **Question Bank** ❖

Q. 1 Determine which machine learning method (Supervised, Unsupervised, or Reinforcement Learning) is suitable for various use cases:

a) **Email Spam Detection**

Which machine learning method would you use to train a model that identifies whether an email is spam or not, using labelled examples?

b) **Customer Segmentation**

If you want to group customers based on purchasing behaviour without prior labels, which machine learning approach is appropriate?

c) **Game AI Development**

What type of machine learning method would you use to train an AI agent to play a game and improve its strategy through trial and error?

d) **Image Classification**

Which method would be suitable for

developing a model that categorizes images into predefined classes using labelled training data?

e) **Market Basket Analysis**

What machine learning approach is best for discovering patterns in customer purchase behaviour, such as which items are frequently bought together?

f) **Recommendation Systems**

If you aim to recommend products to users based on their previous interactions and preferences, which method would you employ?

g) **Anomaly Detection**

What type of machine learning technique would you use to identify unusual patterns in network traffic without labelled instances?

h) **Language Translation**

Which machine learning method

would you use to build a model that translates text from one language to another using large datasets of parallel text?

i) **Robotics Navigation**

What type of machine learning approach would you utilize to train a robot to navigate through an environment by receiving rewards for successful actions?

j) **Disease Prediction**

If you want to predict whether a patient has a certain disease based on labelled medical records, which machine learning method is most appropriate?

Q. 2 Determine which machine learning method (classification or regression) is suitable for various use cases:

a) **Email Filtering**

What type of machine learning method would you use to classify emails as "spam" or "not spam"?

b) **House Price Prediction**

Which machine learning method is appropriate for predicting the selling price of a house based on its features?

c) **Customer Churn Prediction**

If you want to determine whether a customer will leave a subscription service, what method would you use?

d) **Stock Price Forecasting**

Which machine learning approach would be best for predicting the future price of a stock ?

e) **Sentiment Analysis**

What method would you employ to classify movie reviews as "positive," "negative," or "neutral"?

f) **Credit Scoring**

What type of machine learning

technique is suitable for predicting whether a loan applicant is likely to default?

g) **Weather Forecasting**

If you aim to predict tomorrow's temperature based on historical data, which method would you use?

h) **Medical Diagnosis**

What machine learning approach would be effective for diagnosing whether a patient has a particular disease based on their symptoms?

i) **Sales Forecasting**

Which machine learning method would you choose to estimate future sales revenue based on past sales data?

j) **Image Recognition**

What type of method would you utilize to classify images of animals as "cats," "dogs," or "birds"?

Q.3 Give the difference between Supervised Learning and Unsupervised Learning.

Q.4 Give the difference between classification and regression.

Q.5 Give the difference between simple linear regression and logistic regression.

Q.6 Define Machine Learning. Also, compare it with Human Learning.

Q.7 Explain logistic regression with advantage and disadvantage.

Q.8 Write a short note on Single Linear Regression. Also, state applications of it.

Q.9 Discuss the SVM model in detail with its pros and cons.

Q.10 Identify main three features for following well-posed problem:

a. Fruit prediction Problem

b. Handwriting recognition Problem

Q.11 Explain the concept of penalty and rewards in reinforcement learning.

Q.12 Write a short note on learning from examples.

Q.13 Explain types of data in Machine Learning.

Q.14 Give the difference between qualitative data and quantitative data.

• State following statements are whether True or False.

1. **True or False :** Supervised learning requires labelled data for training the model.
2. **True or False :** Unsupervised learning can be used to identify hidden patterns in data without any labelled outcomes.
3. **True or False :** Reinforcement learning involves learning by receiving rewards or penalties based on actions taken in an environment.
4. **True or False :** In supervised learning, the model is trained on input-output pairs.

5. **True or False :** Unsupervised learning is used primarily for classification tasks.

6. **True or False :** Reinforcement learning is often used in scenarios where an agent interacts with an environment to maximize cumulative rewards.

7. **True or False :** Linear regression is used for predicting categorical outcomes.

8. **True or False :** Logistic regression can be used for binary classification tasks.

9. **True or False :** Support Vector Machines (SVM) are primarily used for regression tasks.

10. **True or False :** All three models—linear regression, logistic regression, and SVM—can be used in supervised learning contexts.

❖ MCQ ❖

1. What is the primary goal of machine learning?

- a) To write complex algorithms
- b) To enable computers to learn from data and make predictions
- c) To replace traditional programming
- d) To store large datasets

Answer : b) To enable computers to learn from data and make predictions

2. In supervised learning, the model learns from:

- a) Unlabelled data
- b) Labelled data
- c) Reinforcement signals
- d) None of the above

Answer: b) Labelled data

3. What is the main characteristic of unsupervised learning ?

- a) It requires labelled data
- b) It focuses on finding patterns in unlabelled data
- c) It uses reinforcement signals to learn
- d) It is only applicable to classification problems

Answer: b) It focuses on finding patterns in unlabelled data

4. In reinforcement learning, an agent learns by:

- a) Analyzing historical data
- b) Receiving feedback from its actions
- c) Being provided with labelled examples
- d) Following predefined rules

Answer : b) Receiving feedback from its actions

5. Which of the following is an example of supervised learning?

- a) Clustering customers based on purchasing behaviour
- b) Predicting house prices based on features
- c) Playing a game by trial and error
- d) Reducing the dimensionality of data

Answer: b) Predicting house prices based on features

6. What type of problem is typically addressed by unsupervised learning?

- a) Image classification
- b) Fraud detection
- c) Customer segmentation
- d) Speech recognition

Answer: c) Customer segmentation

7. Which of the following is a common application of reinforcement learning?

- a) Image recognition
- b) Speech synthesis
- c) Game playing (e.g., AlphaGo)
- d) Spam detection

Answer: c) Game playing (e.g., AlphaGo)

8. What is the primary difference between supervised and unsupervised learning?

- a) The amount of data used
- b) The type of algorithms employed
- c) The presence or absence of labelled data
- d) The complexity of the models

Answer: c) The presence or absence of labelled data

9. Which of the following is a common assumption made in linear regression?

- a) The relationship between variables is

nonlinear

- b) The residuals are normally distributed
- c) The output is categorical
- d) All features are independent

Answer: b) The residuals are normally distributed

10. What is the primary purpose of logistic regression?

- a) To predict continuous values
- b) To perform binary classification
- c) To cluster data points
- d) To perform feature selection

Answer: b) To perform binary classification

11. Support Vector Machines (SVM) are primarily used for:

- a) Dimensionality reduction
- b) Clustering
- c) Classification tasks
- d) Regression tasks

Answer: c) Classification tasks

12. What is the key concept behind SVM?

- a) Minimizing the mean squared error
- b) Maximizing the margin between classes
- c) Reducing the dimensionality of data
- d) Finding the closest neighbours

Answer: b) Maximizing the margin between classes

13. Which of the following evaluation metrics is commonly used for classification problems?

- a) Mean Squared Error (MSE)
- b) Accuracy
- c) R-squared
- d) Mean Absolute Error (MAE)

Answer: b) Accuracy

14. In machine learning, what does overfitting refer to?

- a) A model that generalizes well to new data
- b) A model that learns the noise in the training data instead of the underlying pattern
- c) A model that is too simple to capture the underlying pattern
- d) A model that performs well on unseen data

Answer : b) A model that learns the noise in the training data instead of the underlying pattern

15. What technique is commonly used to prevent overfitting ?

- a) Increasing model complexity
- b) Cross-validation
- c) Ignoring irrelevant features
- d) Using more training data only

Answer: b) Cross-validation

16. Which of the following methods can be used for feature selection?

- a) Principal Component Analysis (PCA)
- b) K-means clustering
- c) Linear regression
- d) Gradient descent

Answer: a) Principal Component Analysis (PCA)

17. Which type of model is typically faster to train : a linear model or a non-linear model ?

- a) Linear model

b) Non-linear model

c) They take the same time

d) It depends on the dataset

Answer: a) Linear model

18. Which of the following is a characteristic of reinforcement learning?

- a) It learns from a static dataset
- b) It continuously interacts with the environment
- c) It does not require feedback
- d) It is only applicable to supervised tasks

Answer : b) It continuously interacts with the environment

19. What is a key advantage of using support vector machines ?

- a) They are simple to implement
- b) They work well with high-dimensional data
- c) They require less training data
- d) They are faster than other models

Answer : b) They work well with high-dimensional data

20. In machine learning, the term "hyperparameter" refers to:

- a) Parameters learned by the model during training
- b) The settings that are configured before training the model
- c) Parameters that remain constant throughout the process
- d) Data points used for evaluation

Answer : b) The settings that are configured before training the model

❖ Interview Oriented Questions ❖

1. What is machine learning, and how does it differ from traditional programming ?
2. What is meant by "learning from examples" ?
3. Explain the differences between supervised learning, unsupervised learning, and reinforcement learning.
4. Can you provide real-world examples of each form of learning ?
5. In what scenarios would you choose supervised learning over unsupervised learning ?
6. What are the main goals of reinforcement learning ?
7. What is supervised learning, and how does it work ?
8. Describe the process of training a supervised learning model.
9. What are some common metrics used to evaluate supervised learning models?
10. What is unsupervised learning, and what types of problems does it address?
11. Explain the concept of clustering in unsupervised learning.
12. What is reinforcement learning? How does it differ from supervised learning?
13. Can you explain the concepts of agents, environments, and rewards in reinforcement learning?
14. What challenges might arise in training reinforcement learning models?
15. What is linear regression, and when would you use it?
16. Explain the difference between linear regression and logistic regression.
17. How does logistic regression work?
18. What are Support Vector Machines (SVM), and what are their key advantages ?
19. Can you describe the process of finding the optimal hyperplane in SVM ?
20. How would you interpret the coefficients in a linear regression model ?

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